

Project Logbook

Project Title:
Online Course Popularity & Rating Analysis

Team Members:

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Entry 1: Project Definition

Date: 7 February 2026

Task: Define research focus and measurable variables

Description:

At the beginning of the project, we planned to analyze course popularity using rating as the main indicator. We explored the possibility of collecting rating data from different platforms and successfully scraped rating fields during preliminary testing.

However, after reviewing the collected data, we found that many courses had zero ratings or were marked as not rated. The rating values showed very limited variation, which made them unsuitable for meaningful analysis.

This suggested low student engagement in submitting course evaluations. Therefore, rating was excluded as a primary variable.

We redefined the project focus to analyze measurable enrollment indicators. The number of enrolled students was selected as the main variable, and duration, price, old price, and discount percentage were identified as potential influencing factors.

Output:

Finalized research focus on factors influencing student enrollment rather than rating analysis.

Challenges:

Initial ambiguity in defining a reliable measure of course popularity.

How we handled it:

Shifted from rating-based analysis to structured enrollment and pricing attributes.

Entry 2: Data Source Exploration

Date: 8-9 February 2026

Task: Identify suitable data collection method

Description:

We searched for online learning platforms that provide structured course data. Our initial goal was to use a free public API for reliable and structured extraction.

However, no suitable free API was available. Some platforms required paid access, while others restricted API usage or did not provide APIs at all.

In addition, several websites relied on dynamic JavaScript rendering, which made direct data extraction more complex.

After evaluating accessibility and available attributes, we selected the Shaguf platform because it provides publicly visible course listings that include enrollment, pricing, and duration information.

Output:

Decision to proceed with web scraping as the primary data collection method.

Challenges:

- No suitable free API available
- Dynamic JavaScript-based page structures
- Limited reliability of rating data

How we handled it:

Selected a platform with accessible listing pages and implemented controlled web scraping.

Entry 3: Data Collection Implementation

Date: 10-11-12 February 2026

Task: Implement scraping process in Google Colab

Description:

We implemented a Python-based web scraping script using Google Colab as the development environment. The script retrieved HTML content from multiple course listing pages on the Shaguf platform.

Pagination handling was implemented to ensure collection from multiple listing pages. Controlled request timing was applied to avoid excessive server load and maintain ethical scraping practices.

The following attributes were extracted:

- Course title
- Duration
- Current price
- Old price
- Discount percentage
- Number of students
- Course level
- Instructor
- University
- Rating

Numeric-related fields such as duration and pricing were extracted in their original textual format to preserve raw data integrity.

Output:

Raw dataset exported and stored as a CSV file.

Challenges:

- Extracting numeric values embedded within text strings

- Handling pagination across multiple pages
- Inconsistent formatting of discount and old price fields
- Rating values were successfully extracted; however, most courses had zero or very low ratings, resulting in limited variability.

How we handled it:

Adjusted parsing logic within the scraping script and manually verified sample records for consistency.

Since rating distribution was highly sparse, it was not selected as a primary analytical variable.

Entry 4: Data Review and Storage

Date: 13-14 February 2026

Task: Inspect and organize collected dataset

Description:

After completing the scraping process, the dataset was reviewed to ensure all targeted attributes were successfully extracted. Basic inspection was performed to verify the presence of key fields and check for missing values.

The dataset contains:

- numeric-related variables (price, old_price, duration, discount percentage, students)
- Categorical variables (level, instructor, university)
- Textual variables (title)

The dataset was saved in CSV format without analytical transformation to preserve its raw structure.

Output:

Final raw dataset ready for submission.

Entry 5: additional Data Collection

Date: 2 March 2026

Description: After completing our primary data collection, we needed additional data to support our project. Instead of scraping more websites, we decided to look for ready-to-use datasets on Kaggle.

We searched for online courses and downloaded two datasets from kaggle website:

1. Udemy Courses Dataset
2. Coursera Course Dataset

Since this is secondary data, our main focus was making sure it fits with our primary data. We had to carefully check the columns in these Kaggle datasets to see if they match the attributes we already collected.

Challenges:

- Finding secondary datasets that match the similar columns of our primary data.

Entry 6: Data Processing and Cleaning Phase

Date: 1-3 March 2026

Task: Clean and preprocessing collected dataset.

1. Initial Data Inspection

What was done: Loaded and explored the dataset structure.

Why: To identify data quality issues before cleaning.

How: Used pandas to examine shape, data types, missing values, and summary statistics.

Tools used: pandas, numpy

2. Handling Missing Values

What was done: Removed columns with >40% missing data and relate remaining values.

Why: Excessive missing data introduces bias; remaining gaps need proper handling.

How: Applied 40% threshold for column removal; extracted course codes from titles using regex; filled remaining with "Unknown".

Tools used: pandas, re

Actions taken:

- Removed 8 columns with >40% missing
- Related 190 missing course codes

3. Removing Duplicates

What was done: Removed duplicate records based on URL.

Why: Duplicates represent the same course and skew analysis.

How: Used pandas duplicated () on 'url' column, kept first occurrence.

Tools used: pandas

Actions taken:

- Removed 1 duplicate row with server error
- No loss of valuable information

4. Cleaning Inconsistent Data

What was done: Standardized text and cleaned numerical formatting.

Why: Inconsistent formatting prevents proper analysis.

How: Applied regex to extract numbers; standardized case; normalized university names.

Tools used: pandas, re

Actions taken:

- Cleaned price columns (removed "،" رس fixed commas)
- Standardized university names
- Converted boolean fields to True/False

5. Outlier Treatment

What was done: Capped extreme outliers in students column.

Why: Values up to 25M are data entry errors that cause distortion analysis.

How: Applied capping at 99th percentile (1,103 students).

Tools used: pandas, numpy

Actions taken:

- Capped 6 extreme values (>1M) at 1,103
- Preserved 99% of distribution

6. Data Type Fixing

What was done: Converted columns to appropriate data types.

Why: Proper types enable correct analysis and reduce memory.

How: Used astype() for numerical and categorical conversion.

Tools used: pandas

Actions taken.

- Converted prices to float32
- Converted categorical columns to category type

7. Feature Scaling / Encoding

What was done: Normalized numerical features and encoded categorical variables.

Why: Different scales affect algorithms; categorical data needs numerical representation.

How: Used MinMaxScaler (0-1 range) and one-hot encoding.

Tools used: sklearn.preprocessing, pandas

Actions taken:

- Normalized price, students, duration
- One-hot encoded content_type and instructor_popularity

8. Final Dataset Summary

What was done: Compiled final statistics and saved cleaned data.

Why: To document transformation and ensure reproducibility.

Tools used: pandas

Results:

Metric Value

Original rows 961

Final rows 961

Original columns 29

Final columns 21

Files saved: shaguf_cleaned_final.csv

Entry 7: Exploratory Data Analysis (EDA)

Date: 3-4 March 2026

Task: Perform Exploratory Data Analysis on Primary (Shaguf) and Secondary (Udemy & Coursera) datasets.

1. Primary Factor Analysis (Shaguf Platform)

What was done: Analyzed the relationship between course attributes (price, discount, duration) and student enrollment.

Why: To determine which numerical factors most significantly raise student numbers on the Shaguf platform.

How: Used correlation matrices, scatter plots, and categorical grouping (like binning discounts and durations) to identify trends.

Tools used: pandas, seaborn, matplotlib.

Findings: Price & Duration: Neither price nor course length showed a significant correlation with enrollment, showing that students prefer and prioritize value over cost or time.

- Discounts: Higher discounts did not equate to more students; courses with modest discounts (<50%) showed more students numbers.
- Paid vs. Free: Paid courses attracted more students than free ones, showing students trust in paid content.

2. Instructor Reputation Impact

What was done: Segmented data by instructor popularity levels (Established, Popular, Emerging).

Why: To test if the "instructor reputation" is more important than numerical data like price.

How: Created bar charts comparing the average number of students across different instructor tiers.

Findings:

- Popular instructors have 3.5x more students than emerging instructors, regardless of price or discount. This was a discovery of ours as the primary driver of success on the platform.

3. Visualizing Data Distributions

What was done: Generated histograms and box plots for all key numerical variables (Price, Students, Rating).

Why: To identify the "shape" of the data and detect any remaining anomalies after cleaning.

How: Utilized `sns.histplot` and `sns.boxplot` for single analysis.

Findings:

- Most variables followed a long-tail distribution, with a small number of "popular" courses or instructors getting the most of the market share.
- Anomalies: Discovered that high-discount courses usually had lower enrollment, which is opposite to the normal logic.
- Key Insight: Numerical factors (like price, duration) are not enough to predict success; enrollment from what we observed on shaguf platform depends highly on instructor reputation.

Files used: shaguf_cleaned_final.csv, udemy_courses.csv, coursera_data.csv.

Entry 8: Cross-Platform Comparison

Date: 4-5 March 2026

Task: Comparative analysis of Shaguf (Primary) against Udemy and Coursera (Secondary).

1. Price Benchmarking (Local vs. Global)

What was done: Compared Shaguf course prices (SAR) with Udemy prices converted to SAR.

Why: To evaluate Shaguf's competitive place in the local market compared to globals like Udemy and Coursera.

How: Used Kernel Density Estimate (KDE) plots to visualize the distribution of prices over platforms.

Tools used: pandas, seaborn, matplotlib.

Findings:

- Shaguf has a very specific pricing range for local students, while global platforms show much more variety.

2. Enrollment Analysis

What was done: Analyzed student enrollment numbers across Shaguf, Udemy, and Coursera.

Why: To understand the reach of the platforms and compare their "popularity".

How: compared the numbers of students enrollment across the different platforms.

Tools used: sns.kdeplot, pandas.

Findings:

- Shaguf has high density in specific student ranges, meaning its courses are consistently popular locally. Udemy has a wider spread, showing more variety in how many students join each course.

3. Duration vs price

What was done: did a normalized comparison for duration vs price.

Why: To compare the patterns and correlations directly.

How: normalized the duration and price values for Udemy to a scale of 0 to 1, matching Shaguf's data format.

Tools used: pandas, seaborn, matplotlib.

Actions taken:

- Normalized price values.
- Normalized duration prices.

Findings:

- it revealed that there is a consistent lack of correlation between course duration and price across both local (Shaguf) and global (Udemy) platforms.

3. Course Difficulty Level: Local vs Global

What was done: We used the detailed level data from Shaguf to compare its market place against Udemy and Coursera.

Why: To see if the local platform serves a different difficulty level in comparison to global standards.

Findings:

- We observed that Shaguf is more balanced across 'Beginner' and 'Intermediate' levels in comparison to Udemy, which is heavily dominated by 'Beginner' courses.

Files saved: udemy_cleaned.csv, coursera_cleaned.csv.

Entry 9: Modelling and Model Comparison

Task: Built and compared regression models to predict student enrollment using the combined Shaguf and Udemy dataset.

Model 1: Linear Regression Baseline

Date: 23 April 2026

What was done: Built a Linear Regression model as the baseline model to predict student enrollment.

Why: To establish a simple benchmark model for comparison with more advanced regression models in Phase 3.

How: Used the combined Shaguf and Udemy dataset to ensure consistency across all models. The target variable was `students_log`, and the selected features were `price_log`, `is_free`, and `has_discount`.

Tools used: pandas, numpy, scikit-learn, matplotlib

Actions taken: Applied log transformation to the target variable (`students_capped` → `students_log`) and price (`price_clean` → `price_log`). Created a binary feature (`has_discount`) from the discount rate. Split the data into 80% training and 20% testing. Trained the model using `LinearRegression`. Evaluated the model using MAE, RMSE, and R^2 .

Findings: The Linear Regression baseline model achieved:

MAE = 1.6617, RMSE = 2.0488, and $R^2 = 0.1613$.

The model explained about 16.1% of the variation in student enrollment. The results showed that the model had limited predictive performance. The Actual vs Predicted plot showed scattered predictions, and the Residual Plot indicated that the errors were not fully

random. This suggests that the relationship between course features and enrollment is more complex than what a simple linear model can capture.

Model 2: Random Forest Regressor

Date: 30 April 2026

What was done: Built a Random Forest Regressor model as the second prediction model for the project.

Why: To improve the baseline model and test a more advanced model that can capture more complex relationships between the features and student enrollment.

How: Used the combined Shaguf and Udemy dataset, but rebuilt it to include the duration feature so it matches the research question.

The target variable was `students_log`, and the selected features were `price_log`, `is_free`, `has_discount`, and `duration`.

Tools used: pandas, numpy, scikit-learn, matplotlib.

Actions taken: Prepared and combined the datasets.

Cleaned missing values and converted columns to numeric.

Created new features like `price_log` and `students_log`.

Split the data into 80% training and 20% testing.

Trained the model using `RandomForestRegressor` with `n_estimators = 200` and `max_depth = 6`.

Evaluated the model using MAE, RMSE, and R2.

Findings: The Random Forest model achieved

MAE = 1.79, RMSE = 2.21, and R2 = 0.48.

The model explained about 48% of the variation in student enrollment.

Feature importance: The most important feature was has_discount, followed by price_log and duration, while is_free had the lowest impact.

Challenge: The original combined dataset did not include duration, which is part of the research question.

Actions taken: Rebuilt the combined dataset using both Udemy and Shaguf data and added the duration feature to make the model more complete and consistent with the research question.

Model 3: Decision Tree Regressor

Date: 30 April 2026

What was done: Built a Decision Tree Regressor model as the third prediction model for the project.

Why: To test another regression model that can capture non-linear relationships between course features and student enrollment.

How: Used the combined Shaguf and Udemy dataset to keep the model consistent with the other regression models. The target variable was students_log, and the selected features were price_log, discount_rate_actual, is_free, has_discount, and source_Udemy.

Tools used: pandas, numpy, scikit-learn, matplotlib.

Actions taken: Converted boolean columns into numeric values. Encoded the source platform column. Split the data into 80% training and 20% testing. Trained the model using DecisionTreeRegressor with max_depth = 5. Evaluated the model using MAE, RMSE, and R².

Findings: The Decision Tree model achieved

MAE = 1.6235, RMSE = 2.0223, and R² = 0.1828.

The model explained about 18.3% of the variation in student enrollment.

Feature importance: The most important feature was price_log, followed by discount_rate_actual.

Challenge: At first, the Decision Tree model was built using only the Shaguf dataset, while the other models used the combined dataset.

Action taken: Re-trained the Decision Tree model using the same combined Shaguf and Udemy dataset to make the comparison fair.

Model Comparison and Best Model Selection

Date: 1 May 2026

What was done: Compared the performance of Linear Regression, Decision Tree Regressor, and Random Forest Regressor.

Why: To identify which regression model performed best for predicting student enrollment.

How: Compared the models using MAE, RMSE, and R^2 . All models were evaluated using the combined Shaguf and Udemy dataset to keep the comparison fair.

Tools used: pandas, matplotlib.

Results:

Model	MAE	RMSE	R^2
Linear Regression	1.6617	2.0488	0.1613
Decision Tree	1.6235	2.0223	0.1828
Random Forest	1.7900	2.2100	0.4800

Findings: Decision Tree had the lowest MAE and RMSE, meaning it had the smallest prediction error. However, Random Forest achieved the highest R^2 score, meaning it explained the largest part of the variation in student enrollment.

Final decision: Random Forest was selected as the best overall model because it had the highest R^2 score and captured the general pattern in the data better than the other models.

Files saved: Decision_Tree_Model.ipynb Phase3_Model_Comparison.ipynb

References:

- Kaggle website :
<https://www.kaggle.com/>
- Udemmy Courses Dataset:
<https://www.kaggle.com/datasets/refiaozturk/udemmy-courses-dataset>
- Coursera Course Dataset:
<https://www.kaggle.com/datasets/siddharthm1698/coursera-course-dataset>